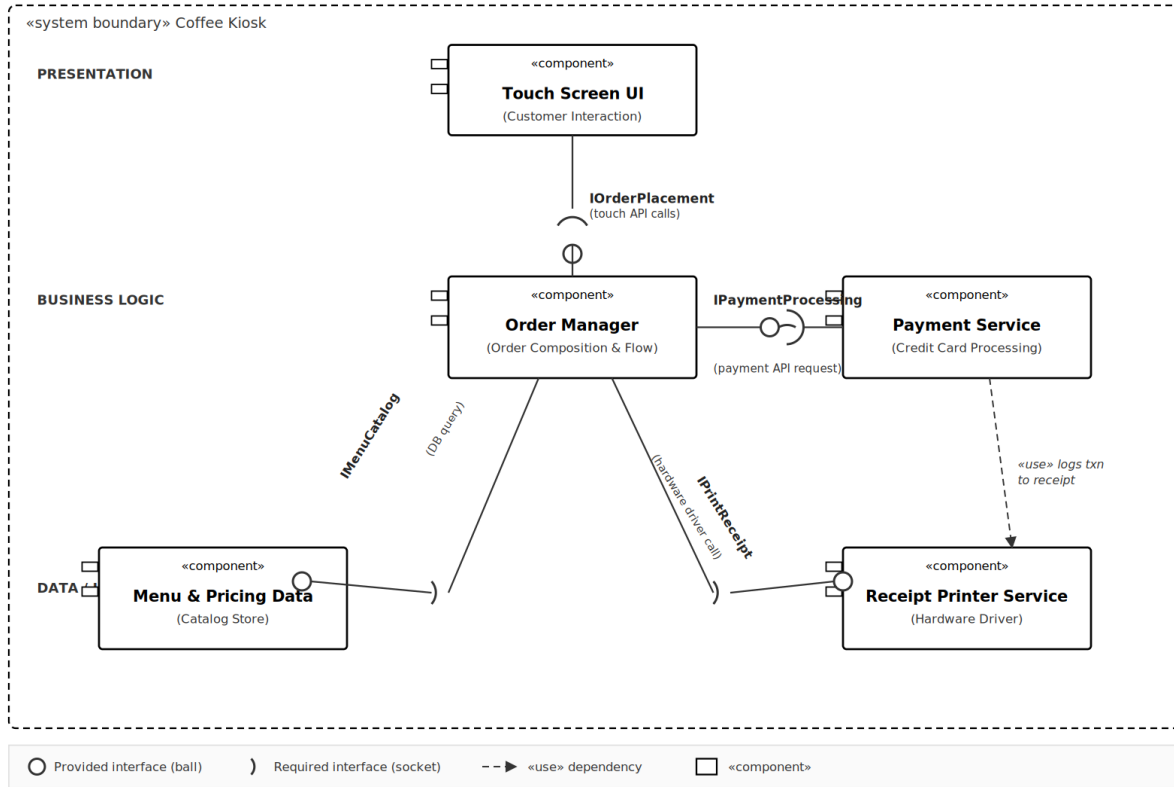


Lab 3: Component Diagram & Architectural Justification

Self-Service Coffee Kiosk System

Self-Service Coffee Kiosk System — UML Component Diagram

Layered Architecture



Architecture Selection: *"We chose Layered Architecture for the Self-Service Coffee Kiosk System."*

Architectural Choice

A four-layer architecture was selected: a Presentation layer (Touch Screen UI), a Business Logic layer (Order Manager, Payment Service), a Data layer (Menu & Pricing Data), and a Hardware Interface layer (Receipt Printer Service). Each layer only calls the layer directly below it through a well-defined interface, keeping the customer-facing screen isolated from payment logic, catalog data, and printer hardware.

Two Scenario-Related Reasons

- **Single, self-contained device:** The kiosk is one physical unit with a fixed, small scope (3 coffee types, 2 sizes, one payment method), so there is no independent-scaling or multi-team-deployment need that would justify Microservices' operational overhead. A Layered design gives the same clear separation of concerns with far less complexity.
- **Straightforward hardware and data integration:** The touch screen, receipt printer, and menu/pricing store are all local to the kiosk. Layering lets the Order Manager mediate cleanly between the UI, the printer driver, and the pricing data without exposing hardware details to the presentation layer.

Security Advantage

Because the Touch Screen UI can only reach the Payment Service indirectly through the Order Manager's IOrderPlacement/IPaymentProcessing interfaces, raw card data never has to pass through or be exposed to the presentation layer. This layer isolation narrows the attack surface for card-data handling and keeps PCI-relevant logic confined to a single, auditable layer.

Performance Benefit

Menu and pricing data is read directly by the Business Logic layer from a local Menu & Pricing Data component rather than over a network call (as a Microservices split might require), so order totals and item lookups are computed with in-process, low-latency access, keeping the touch-to-confirmation time short for customers standing at the kiosk.